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## identity model

→ VCG is DIC

→ Pivot:

① no negative externalities

$$v_i(d, \theta_i) \geq 0$$

② no single agent effect

⇒ weakly budget balanced

① no negative externalities

② outcome set is monotonic

⇒ ex-post I.R.

VCG inefficient as # of objects increases

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Example 8:

Buyer:  $\theta_B$  | Suppose  $\theta_B > \theta_S \Rightarrow$  Allocates to buyer, who  
Seller:  $\theta_S$  | pays  $\theta_S$ . Seller receives  $\theta_B$

Myerson-Satterthwaite Thm:

Thm 6.9:

$$\theta_S: F_S \rightarrow [a_S, b_S]$$

$$\theta_B: F_B \rightarrow [a_B, b_B]$$

overlap of intervals  


if  $\Pr([a_S, b_S] \cap [a_B, b_B]) > 0$ ,  $\exists$  no mechanism  
which is efficient, BIC, budget balanced, I-IR

Proof: Do full by self.

Satisfy 3 ~~4~~  $\times$  4.  $\times$  EFF, BIC, BB

Proof:  $x(\theta_s, \theta_B) := \Pr(\text{buyer gets item})$ ,  $\bar{x}_s(\theta_s) = E_{\theta_B}[x(\theta_s, \theta_B)]$ ,  $\bar{x}_B = E_{\theta_s}[x(\theta_s, \theta_B)] := \Pr(\text{each get object, respectively})$ .

$\bar{t}_i(\theta_i) = E_{\theta_{-i}}[t_i(\theta_i, \theta_{-i})]$  for  $i \in \{S, B\} :=$  expected payment.  $U_s(\theta_s) = \bar{t}_s(\theta_s) - \theta_s \bar{x}_s(\theta_s) :=$  expected payoff to seller.  $U_B(\theta_B) = \theta_B \bar{x}_B(\theta_B) - \bar{t}_B(\theta_B) :=$  " " " " buyer.

$$\left. \begin{array}{l} \text{Buyer: } U_B(\theta_B) = u_B(a_B) + \int_{a_B}^{\theta_B} \bar{x}_B(k) dk \\ \bar{x}_B(k) \text{ is non-decreasing} \\ \text{Seller: } U_s(\theta_s) = u_s(b_s) + \int_{\theta_s}^{b_s} \bar{x}_s(k) dk \\ \bar{x}_s(k) \text{ is non-increasing} \end{array} \right\} \mathcal{IC}$$

↳ buyer receives

Budget Balance:  $t_s(\theta_s, \theta_B) + t_B(\theta_s, \theta_B) = 0$

$$\int_{a_B}^{b_B} \int_{a_s}^{b_s} \underbrace{(\theta_B - \theta_s) x(\theta_s, \theta_B)}_{\text{↳ } E[\text{payoff}]} \underbrace{f_B(\theta_B) f_s(\theta_s)}_{\text{↳ density } f(\cdot)} d\theta_s d\theta_B$$

$$= \int_{a_B}^{b_B} \int_{a_s}^{b_s} [(\theta_B - \theta_s) x(\theta_s, \theta_B) + t_s(\cdot) - t_B(\cdot)] f_B(\theta_B) f_s(\theta_s) d\theta_B d\theta_s$$

$$= \int_{a_B}^{b_B} \int_{a_s}^{b_s} [u_B(\cdot) + u_s(\cdot)] f_B(\theta_B) f_s(\theta_s) d\theta_s d\theta_B$$

$$= \int_{a_s}^{b_s} U_s(\theta_s) f(\theta_s) d\theta_s + \int_{a_B}^{b_B} U_B(\theta_B) f_B(\theta_B) d\theta_B$$

↳ Exp. payoffs. Int by parts to get following

$$= U_S(b_S) + U_B(a_B) + \int_{a_B}^{b_B} \int_{a_S}^{b_S} F_S(\theta_S) f_B(\theta_B) + (1 - F_B(\theta_B)) f_S(\theta_S) f_B(\theta_B) d\theta_S d\theta_B$$

↳ find via int. by parts. Written in notes on eLC

$$U_S(b_S) + U_B(a_B) = - \int_{a_B}^{b_S} (1 - F_B(\theta_B)) f_S(\theta_B) d\theta_B$$

if  $\Pr([a_S, b_S] \cap [a_B, b_B]) > 0$ ,  $a_B > b_S$ ,  $a_S < b_B$

$$\Rightarrow - \int_{a_B}^{b_S} (1 - F_B(\theta_B)) f_S(\theta_B) d\theta_B < 0 \quad \times$$

violates IR of buyer or seller

showing  $\sum u_i(\cdot) < 0$ .